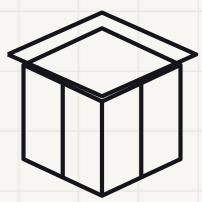

CRATES

A scarce registry for agent-bearing collections on Bittensor EVM

128 numbered seats. Every crate is a curated container of on-chain agents — measured by settled work and verified volume, never by hashpower. Subnets taught the lesson; Crates apply it to digital property.

5

FABLE

Infinite listings are worth nothing

Every NFT marketplace made the same choice: list everything, curate nothing. The result is a market where being “on the marketplace” carries no information at all. Bittensor made the opposite choice for machine intelligence — a hard cap of **128 subnet slots**, priced by demand, pruned by performance — and the seat itself became one of the most contested assets in crypto. **Crates** apply that discipline to the one asset class that can now be measured the way subnets are measured: collections of on-chain agents that do paid, settled, auditable work.

The thesis

A **Crate** is a numbered registry seat — `#001 … #128` — that binds a collection into the 8004 protocol: its contracts, its manifest, its markets, its provenance ledger, and above all its **agents**. On 8004, an NFT is not a picture; it is an economic actor with an identity, a token-bound account, a work history, and — uniquely on chain 964 — a substrate citizenship: its account can own a Bittensor neuron. A Crate is the container that makes a set of those actors *legible, comparable, and scarce*. Registries have been tried before and have died before; §02 is honest about why, and §03 states the one thing that changed.

128 CRATE SEATS — THE CAP BITTENSOR CHOSE FOR SUBNETS	**3** SEATS FILLED TODAY (2 LIVE · 1 SEALED) · 2 REJECTIONS ON THE RECORD	**τ 632** VERIFIED ALL-TIME VOLUME ACROSS LIVE CRATES, INDEXED FROM CHAIN	**5** REGISTERED AGENTS · 7 ESCROWED TASKS SETTLED · τ 0.018 PAID TO AGENT ACCOUNTS

How to read this paper

This registry's product is honesty, so the paper is typed the way the platform is built. Every capability claim carries one of three marks, and the marks are load-bearing:

PROVEN Fired live on chain 964; the receipt (tx, block) exists and is cited. Nothing inferred, nothing rehearsed on a testnet.	DESIGNED A concrete construction, buildable on chain 964's actual runtime today, with named standards and named dependencies. Not yet fired.	SPECULATIVE Depends on infrastructure or standards the chain may not have. Stated so a reader can see the edge of the map.

Where this paper says *proven*, an appendix row in §13 gives the transaction. Where the honest answer is “not yet” — and there is one load-bearing *not yet* in §03 — the paper says not yet. Chain is the auditor, even in prose.

> **What Crates deliberately are not.** No mining. No hashpower. No emissions schedule. No crate tokens — not now, not at any phase. The measurable substance inside a Crate is commerce and agent work that already settles on chain 964. TAO is the only money in the system, and the ledger is the only oracle.

Registries die of bad measurement

Curated registries are not a new idea. They are a repeatedly *failed* idea, and a paper proposing one owes the reader the autopsy before the pitch.

The token-curated registry, 2018

TCRs proposed exactly this shape: a list worth being on, guarded by economic stakes. Token holders voted entries in and out; the token's value was supposed to track the list's credibility. In practice the flagship (adChain, 2018) and its imitators died within a couple of years, and the failure was structural, not operational. **Votes aren't facts.** When admission turns on a subjective judgment — *is this a quality entry?* — the vote measures the voters, not the entry: apathy, reflexive token-price games, and whale capture do the rest. The design literature said the quiet part itself: TCRs work in proportion to how *objective* the listing criterion is. Almost none were.

The survivors, and what they prove

Two curation designs outlived the graveyard, and both prove the same rule from opposite ends. **Arbitration-backed lists** (the Kleros line) survive by making subjectivity explicit and charging for it — every dispute goes to a paid court. **Marketplace checkmarks** survive by costing nothing and claiming almost nothing — verification on an infinite-shelf venue is anti-fraud plumbing, not curation; it cannot rank, only exclude. The rule both prove: **a registry can only be as honest as its measurement.** Either you buy honest judgment (courts), or you shrink the claim until no judgment is needed (checkmarks) — or you find a category where the measurement already exists as fact.

The physical-world control case

Capped registries exist off-chain too, and the largest one is a warning label. The taxi medallion was a numbered, transferable, capped seat — and seven-figure medallion prices collapsed when the cap stopped binding the category. The lesson Crates takes seriously rather than around: **a cap prices access to a category; it cannot create the category's worth.** The 128 seats are only as valuable as agent-borne commerce on chain 964 becomes. This paper bets on that category and says so plainly (§13 names it as the first risk).

> **The bar this history sets.** A registry that wants to outlive the graveyard needs all three: an **objective criterion** (facts, not votes) · a **binding cap** (allocation, not opinion) · a **measurement that exists independently of the registry** (so the registry can be wrong and be caught). TCRs had the cap and nothing else. Checkmarks have facts and no cap. Bittensor's subnet registry is the first system with all three — and its criterion, emissions, only exists because miners exist. §03 is about the moment a second category acquired all three *without* miners.

Property that can work

The measurement problem dissolved quietly, in three steps, and the third one happened on chain 964 specifically. An NFT stopped being a picture and became an economic actor whose work leaves receipts.

Step one — the account PROVEN

ERC-6551 gives every token a deterministic smart account — a **token-bound account** whose address never changes when the NFT is sold. On 964 this is live: five registered agents hold TBAs resolved from the deployed registry, and one of them has held, staked, and been paid TAO. Whatever the account earns, owns, or is attested to **survives the sale by construction** — the address is the constant; §09 builds on exactly this property.

Step two — the identity and the work PROVEN

An ERC-8004-family identity registry (bespoke implementation, live on 964 since June 2026) binds token → agentId → account, and an escrowed task registry gives the agent a **work history that costs money to fake**: a requester escrows TAO at creation; settlement pays the agent's TBA; every state transition is an event. Seven tasks have run the full lifecycle live — settle, refund, and a permissionless settle fired 9.8 hours after the window opened, by receipt. Reputation built from these rows is reputation someone paid for.

Step three — substrate citizenship PROVEN

Chain 964 is Bittensor's own EVM, and it hides a bridge: when a contract account calls a substrate precompile, the runtime derives its substrate identity as `blake2_256("evm:" ‖ address)`. On 2026-06-29 an agent's TBA fired `burnedRegister` through its mandate and became the **coldkey of a live Bittensor neuron** — SN116, UID 86 — and the derived coldkey matched the chain's, byte for byte. An NFT's account owns a miner registration. That is not a metaphor; it is `0xabd928…aedab · blk 8,513,851`.

> **The one unproven link — stated, not buried.** The registration above was made on a deliberately dead subnet: it proves **mechanism and custody**, not economics. **No agent has yet earned a single RAO of mining emission.** Earning on a paying subnet is the platform's next live test, and until it fires, every sentence in this paper about agents *mining* is a sentence about a proven registration path and an unproven revenue path. A whitepaper that blurred that line would be worth less than this page.

Put the three together and the graveyard's bar from §02 is met without a single miner on the registry's own payroll: the criterion is objective (settled TAO, token-verified sales), the cap can bind (128), and the measurement exists **independently** — every input is a public chain event a third party can recount. For the first time, a registry of digital property can score its cargo the way Bittensor scores subnets: **by what it settles.**

What a Crate is

A Crate is not a listing. A listing is a row anyone can add; a Crate is a **seat** — numbered, scarce, stateful, and eventually ownable. Each Crate binds five things:

1 · The Manifest

Identity of the cargo: NFT contract(s), supply, token-id scheme, render mode (raster, procedural-3D, remote), category. Today a registry row; at P1 an on-chain object anchored to the collection's own `contractURI()` with its hash frozen at seating — manifest drift becomes detectable, not deniable (§08).

2 · The Agent Surface

Every token in a crate can register as an agent — identity, token-bound account, mandates, work history. The crate is the census tract for its agents, and on 964 each agent account is also a substrate citizen (§03, §07).

3 · The Market Venue

Native, audited trading (TunkMarket on 964) with contract-verified buys — the displayed price is re-read from the chain at click time. Own-venue collections keep their own markets, honestly labeled as such, mint volume included.

4 · The Provenance Ledger

Every token's history from genesis mint through every sale, reconstructed from the chain archive and exposed transaction-by-transaction. A crate's volume is the sum of rows anyone can audit.

> **5 · The Score.** The measured output of everything above: settled tasks, TAO paid to agent accounts, token-verified volume, active agents, distribution. One number per crate, a pure function of public chain state, published verifiably (§09, §11) — the analog of subnet emissions, without a single miner. DESIGNED

Crate states

STATE	MEANING	TODAY'S EXAMPLE
LIVE	Manifest bound, contracts verified, market and ledger active.	#001 TaoPunks · #002 ALCI
SEALED	Seat reserved, cargo unconfirmed. Rendered as a strapped crate that makes no claims — absent beats invented.	#003 TaoNauts
DORMANT	Live crate whose score has decayed below the floor; challengeable once past immunity (§07, §10).	–
RECLAIMED	Seat recycled to the curve after grace; the lock returns, the history stays on the ledger forever. Recycled, never burned.	–

Curation with teeth — already demonstrated

The registry has already said *no*. Two candidate collections (Chunkies, Taoist) were surveyed, scoped, and then **withdrawn** when their launchpad collapsed and the teams' commitment failed the bar — seats reclaimed before ever going live. A registry that cannot reject cannot curate; this one rejected before it scaled, and the precedent is the cultural core of the system.

Why 128 seats

128 is not aesthetic. It is the number Bittensor converged on for subnet slots — large enough for a real economy of specialization, small enough that every seat stays legible and every admission is contested. The parallel is worth stating precisely, because Crates borrow the mechanism while deleting the machinery.

	BITTENSOR SUBNETS	8004 CRATES
THE SEAT	Netuid, capped at 128 active slots (expanding toward 256 under demand)	Crate number, capped at 128 (`#001–#128`)
THE CARGO	A market for machine intelligence (miners + validators)	A collection of on-chain agents (NFTs with identity, capital, work)
ADMISSION	TAO lock; cost doubles per registration, decays between claims	P2: the same curve — lock, double, decay, refund on exit (§10)
MEASUREMENT	Emissions from validator consensus over miner work	CrateScore from settled tasks, verified volume, agent activity (§11)
PRUNING	Weakest subnet deregistered when slots are full; immunity protects new entrants	Lowest-score crate past immunity becomes challengeable → DORMANT → RECLAIMED (§07)
WHAT'S DELETED	Mining, hashrate, validator hardware, emission schedules, per-seat tokens	— all of it. The chain already settles the work; Crates only measure it.

The scarcity arithmetic

OpenSea hosts millions of collections; the strongest signal a collection can earn there is a checkmark. Under a 128-cap, the *weakest* admitted crate is still one of 128 — top of the category by construction. When admission also locks capital that a stronger entrant can contest, holding a seat becomes a continuous, priced claim: *this cargo is still worth its slot.* Demand for Bittensor's slots pushed subnet registration into four-figure TAO territory by 2026 — none of that value came from the slot doing anything; it came from the slot being one of 128.

The seat as an asset

At P1 the CrateRegistry mints each seat as an ERC-721 — **128 crate tokens, ever**. Owning `CRATE #007` is the right to bind and configure its manifest, subject to protocol rules. Seats transfer like any NFT; the registry doesn't care who curates, only that the cargo performs. This mirrors subnet ownership — and quietly makes the seats the platform's first native scarce asset without issuing a single fungible token. DESIGNED

> **Levelling honestly:** the cap is a governance choice, not a law of nature — Bittensor itself is expanding 128 → 256 under demand. Crates treat 128 as the constitution's opening article: changeable, but only loudly, slowly, and with existing seat-holders' interests priced in.

The machine that already runs

Crates are not a cold start. The centralized version is deployed and operating at **8004.build** — a curated register with three seats filled, backed by indexers that make every number on screen auditable against chain 964.

The register, as of 2026-07-11

SEAT	CARGO	CHAIN FACTS (ALL VERIFIED ON 964)	STATE
#001	TaoPunks	3,333 ERC-721s · verified all-time volume **τ 336** (v1 305.99 + v2 30.36, token-verified sale-by-sale) · **5 registered agents** · full trait & bias-audited rarity system	LIVE
#002	ALCI	First crypto-art collection on subtensor EVM · 286/1,024 minted (recounted from genesis Transfer logs, 2026-07-11) · on-chain metadata, procedural 3D rendered natively · own-contract volume **τ 296** (286τ mint + 10τ secondary), full mint+sale ledger exposed	LIVE
#003	TaoNauts	Seat reserved; cargo unannounced. Rendered as a sealed, strapped crate that claims nothing.	SEALED
#004–128	—	125 seats unclaimed. Under P2 economics, early seats are the cheap ones.	OPEN

Infrastructure proven in production — with receipts

LAYER	PROOF	RECEIPT
IDENTITY	Five agents registered live through the deployed registry, incl. one through the production /register flow end-to-end	`0x5EbB…ce0e · agents 1–5`
CUSTODY	An agent's TBA became a Bittensor neuron coldkey; derivation matched on-chain, byte-exact	`0xabd928…aedab · UID 86`
CONTROL	Scoped mandates drove registration + stake ops through the manager; withdrawal selectors unencodable by construction	`0x4FE4…1703 · 3 executions`
WORK	7 escrowed tasks lifecycle-complete: settle, refund, and a permissionless settle 9.8h post-window — no rejection path post-submit, by receipt	`0x1dC9…8B01 · tasks 1–7`
VOLUME TRUTH	A sale counts only when the venue received TAO and the NFT moved in the same tx; wash-resistant by construction, audited to the wei	`τ336.34 + τ296 = τ632`
NATIVE TRADING	Contract-verified buys: price re-read from `getListing` at open and at click — the UI cannot show a price the chain didn't just say	`0xB546…060D`

Full 64-hex transaction hashes, blocks, and re-verification procedures for every receipt: `docs/verification-anchors.md` in the 8004labs repository (chain-verified 2026-07-11).

> **Honest status:** Phase 0 is centralized. The register is a database table; the curator is 8004 Labs; admission is editorial. Everything *measured* is chain-true, but the *seat allocation* is not yet on-chain. The rest of this paper is the path from here to there.

The full subnet curriculum

The cover says subnets taught the lesson. Most projects that borrow Bittensor's aesthetics stop at the cap. The mechanism has at least six more teachings, and Crates adopts, adapts, or refuses each one explicitly.

SUBNET TEACHING	CRATES APPLICATION	POSTURE
THE LOCK CURVE	Admission priced by demand with no auctioneer: lock doubles per claim, decays toward a floor between claims, refunds on exit (§10).	DESIGNED
IMMUNITY	A freshly bound crate cannot be challenged during its immunity window — new cargo gets time to breathe before the score compares it to incumbents. Imported wholesale from neuron + subnet dereg rules.	DESIGNED
THE CHALLENGE	When 128/128, a claimant's lock doesn't join a queue — it **challenges** the lowest-scoring unprotected crate. Bittensor evicts by emissions; Crates evict by CrateScore. Eviction is the price of a binding cap, and it is the mechanism TCRs never had: the challenger stakes against a *fact*, not a vote.	DESIGNED
THE TEMPO	Subnets re-earn their emissions every epoch; nothing is grandfathered. CrateScore is computed in epochs with recency decay — history earns respect, only activity earns rank (§11).	DESIGNED
RECYCLE ≠ BURN	Bittensor distinguishes TAO recycled (returned to the schedule) from burned (destroyed). Crates inherit the vocabulary as policy: reclaimed seats return to the curve; locks return to claimants; nothing of a crate's history is destroyed — RECLAIMED archives, never erases.	DESIGNED
CUSTODY / OPERATION SPLIT	Bittensor separates coldkey (owns) from hotkey (operates); child-hotkey delegation deepens it. The *derivation* is trustless and proven byte-exact — and the TBA-is-coldkey identity itself is **triangulated by three independent reads agreeing to the RAO** (EVM balance, direct substrate storage, and the 0x80E raw-balance precompile, on two accounts with a control — anchors §13): the TBA IS the coldkey, and a derived coldkey has no keypair — stake can enter only through the owner's own hands, and pure value exits only owner-direct. The *mandate mechanism* is live on 964 (operator drives through a selector-whitelisted mandate). But an adversarial contract review (2026-07-17) found the operator **bounds are app-enforced, not contract-enforced**: MandateManager will whitelist any selector (withdrawal vectors are unencodable *by this app's discipline*, not by the contract), and a re-grant widens rather than narrows while a sale carries the seller's grants to the buyer. A one-contract v2 (swap, no stake migration) closes both. So: **trustless derivation, tiered custody** — and the founding collection (TaoPunks) is **Tier 2**, stated not hidden: it has a burn function and a live admin key, so 8004 makes a best-effort custody promise with that trust assumption named, not an eager guarantee. The third-party split itself has also never been exercised (8 of 8 mandates are owners operating their own agents). An argument is not a proof; a discovered gap even less so.	PROVEN DESIGNED
CONTINUOUS REPRICING (dTAO)	dTAO's deepest lesson: price each seat's flow continuously, in the market, not in governance. Crates adopt the *repricing* and refuse the *token*: the seat NFT + lock curve + challenge is the priced object. If a tradable index of crate activity is ever justified, dTAO itself is the native path — noted, not promised.	SPECULATIVE

The deepest inheritance is the bridge

Everything above is mechanism transplanted. One inheritance is not a transplant but a birthright: through `blake2_256("evm:" ‖ account)`, every agent account in every crate is a first-class substrate citizen. One has already owned a neuron registration and held subnet alpha PROVEN. What follows from citizenship — agents as delegators, agents as subnet participants at scale, substrate-side identity surfaces — is the platform's research horizon SPECULATIVE, and the one revenue claim it implies remains the unproven link named in §03.

Standing on standards, honestly

Crates composes with the agentic-standards frontier instead of inventing beside it. Statuses below were checked against primary sources on 2026-07-11 (✓) or carry their as-of dates; the calibration mark says what a Crates feature built on each could honestly claim *today*.

STANDARD	STATUS	WHAT IT GIVES CRATES	MARK
`ERC-6551 token-bound accounts`	Review ✓ — canonical pattern, stable in practice	The spine. Agent accounts with sale-invariant addresses; our TBA-as-coldkey is a novel application of the standard, not a fork of it.	PROVEN
`ERC-8004 trustless agents`	Draft ✓ — reference deployments live since Jan 2026	The identity vocabulary (bespoke implementation live on 964). Its open-feedback Reputation Registry we deliberately do **not** adopt — §09 cites the evidence.	PROVEN
`ERC-721 the seats`	Final	128 crate tokens ever; seat ownership = curation rights, transferable, no fungible issuance (§05).	DESIGNED
`ERC-7572 contractURI`	Draft ✓ — de-facto marketplace standard	The manifest layer: seats require cargo to self-describe on-chain; manifest hash frozen at seating makes drift detectable. `ContractURIUpdated` feeds the indexer.	DESIGNED
`ERC-5192 soulbound (locked)`	Final ✓	Non-strippable service records minted *to the TBA* — the reputation construction of §09.	DESIGNED
`EAS attestations`	9.5M+ attestations as of 2026-05 ✓ · self-hostable, tokenless	CrateScore epochs as verifiable on-chain attestations with merkle roots over settlement events (§09). No EAS instance exists on 964 yet — we would deploy it.	DESIGNED
`ERC-7656 token-linked services`	Final ✓ (Oct 2025)	The standards-blessed upgrade path for attaching per-seat/per-agent service contracts (manifests, records) at deterministic addresses — noted as the extension route, not built on today.	DESIGNED
`ERC-7715 / 7579 permissions · modular accounts`	Draft / adopted de facto (as of early 2026)	Convergence target for mandates — see below.	SPECULATIVE
`ERC-4337 / EIP-7702 account abstraction`	Live on Ethereum (2023 / Pectra 2025)	Gasless operator UX — but no bundler infrastructure is known on 964 and 7702 tx-type support there is unverified. Named so the edge of the map is visible.	SPECULATIVE
`x402 / AP2 agent payment rails`	Live, stablecoin-first (as of early 2026)	Crate agents could advertise x402-metered endpoints on their agent cards. But x402 *meters requests*; our task registry *escrows outcomes* — and CrateScore counts only what settles on 964. Off-chain receipts never enter the score.	SPECULATIVE

Why mandates already fit the frontier

The MandateManager — per-target selector whitelists, spend limits, expiry windows — is a pre-standard implementation of what session-key standards are converging on. One constraint keeps it load-bearing rather than legacy: at the precompile boundary, **the TBA must be the immediate caller**, because the caller *is* the coldkey.

Bundler-relayed abstraction cannot cross that line; permission layers on 964 must wrap *around* the account, exactly as mandates do. Convergent design, kept by physics, migratable by choice when the standards arrive on-chain.

A record that survives the sale

Agent reputation has a hard question at its core: when the NFT is sold, does the record follow the property or the person? Most designs pick one and lose the other. On this stack the question dissolves — and the industry's first empirical data says the usual alternative fails outright.

The evidence against open feedback

ERC-8004's own answer to reputation is an open feedback registry. A 2026 empirical study of the deployed 8004 ecosystem (arXiv:2606.26028) found that only **3–15%** of registrations across three chains expose a valid registration file with a live endpoint, and that the feedback registry **“cannot function as a trust signal”** — reviews are ungrounded and Sybil reviewers coordinate freely. This is the TCR failure again, one layer down: **feedback that costs nothing measures nothing.** Crates therefore adopts 8004's identity layer and rejects its reputation layer, replacing it with records that cost escrowed TAO to create.

The construction DESIGNED

1 · Settlement mints the record

On `settle()`, a hook mints a **locked ERC-5192 token to the agent's TBA**: taskId, value, task-type, and — critically — **the operator address at settlement time**. Locked means the seller cannot strip it before listing; minted-by-registry means no mandate expansion.

2 · The address is the constant

The TBA address is invariant under NFT transfer (ERC-6551), so the record **travels with the property through every sale, by construction** — no transfer ceremony, no reputation-migration primitive, nothing to forget.

3 · The operator stamp prices the nuance

A buyer sees the full service record and can discount epochs earned under a previous operator. Provenance belongs to the property; performance attribution belongs to whoever held the keys. The market prices the difference; the protocol just refuses to lose the data.

4 · Score epochs become attestations

Each CrateScore epoch is published as an EAS attestation on 964: `(epoch, crateId, score, settledVolume, merkleRoot, prevEpochUID)`. The merkle root commits to the settlement events behind the score; `refUID` chains the epochs. Third parties recompute the root from the task registry's own logs.

> **What this buys.** The leaderboard stops being an indexer artifact and becomes an on-chain series anyone can falsify: recompute the root, walk the chain of epochs, disagree with a pull request instead of a dispute. Reputation earned by paid work, bound to the asset, priced by the market, verified by the ledger — the four properties open-feedback systems have not delivered.

Dependencies, named: an EAS deployment on 964 (plain Solidity, no special runtime — comparable contracts already deployed there without incident); the settle-hook contract; the epoch publisher key, whose authority shrinks over time as the schema resolver enforces roots the task registry itself accumulated. All DESIGNED; none fired. When the first record token mints live, this page gets its receipt.

The lock curve & seat lifecycle

P2 admission borrows Bittensor's battle-tested pricing because it solved exactly this problem: pricing a fixed set of seats under unknown demand, without an auctioneer, without governance in the loop. DESIGNED

```
// seat pricing (per successful claim)
lock_cost(t) = max( FLOOR,
                    last_lock_cost × 2 × decay(t − t_last_claim) )

decay(Δt)    = halves toward FLOOR over DECAY_INTERVAL
               (cost falls back when nobody claims)

on_claim     : TAO locked in CrateRegistry   (escrowed — never burned)
on_exit      : lock returned, seat reopens at current curve price
on_challenge : at 128/128, a claim targets the lowest-CrateScore
               crate past IMMUNITY; incumbent's lock returns,
               manifest archived, history immutable
```

- **Lock, not burn.** Capital is committed, not destroyed — the seat-holder's stake is recoverable by exiting or by losing the challenge, which keeps the claim honest without being punitive.
- **Doubling + decay finds the market price** with no committee: hot demand compounds cost geometrically; cold periods let it fall back toward the floor.
- **Immunity for new cargo.** A freshly bound crate cannot be challenged during its immunity window — the subnet deregistration lesson, imported wholesale (§07).
- **The challenge stakes against a fact.** The challenger doesn't argue the incumbent is unworthy; the score already said it. Capital only decides who occupies the seat the measurement vacated.

Flows, and where they go

FLOW	SOURCE	DESTINATION
SEAT LOCKS	Crate claimants (P2)	Escrowed in CrateRegistry; refundable — never revenue
REGISTRATION FEES	Agent registrations inside crates	Protocol (the existing τ0.01 fee-flip precedent at traction)
VENUE FEES	Native-market trades of crate cargo	Protocol today; P4 option: split with the crate's treasury
AGENT EARNINGS	Settled tasks (escrowed TAO)	The agent's own token-bound account — never the platform

> **The no-token covenant, restated.** Crates issue no fungible token at any phase. TAO is the money; the crate seat (an NFT, 128 ever) is the only new scarce asset, and its value derives from the right to curate — not from a promise of cash flows. Locks are refundable; seats are utility. If the ecosystem ever deserves a tradable index of crate activity, dTAO already demonstrates the native path — a bridge for a different decade of this project, and not promised here.

CrateScore: proof of commerce

Subnets need miners because intelligence must be *generated* before it can be measured. Crate cargo is different: agents and collections already generate settled, priced, public activity on chain 964. The score doesn't create work — it reads it. Every input below is **already indexed in production** by the Phase-0 platform.

AXIS	CHAIN-TRUE INPUT (SOURCE OF TRUTH)	STATUS TODAY
AGENT WORK	Tasks settled by the crate's agents; TAO escrowed → paid to TBAs (task-registry events)	INDEXED · LIVE
AGENT CENSUS	Registered identities per crate; active mandates	INDEXED · LIVE
VERIFIED VOLUME	Token-verified sales (TAO in + NFT moved, same tx) across all venues, mint included	INDEXED · LIVE
CAPITAL AT WORK	TAO and subnet alpha held/staked by the crate's TBAs	INDEXED · LIVE
DISTRIBUTION	Distinct holders; concentration (a 7-wallet collection is not a community, and the score should know it)	DERIVABLE
LIVENESS	Recency decay on all of the above — history earns respect, only activity earns rank (the tempo, §07)	P3 DESIGN

Gaming, honestly confronted

- **Wash trading** → token-verification helps but does not cure self-dealing. Mitigations: volume contributes with diminishing weight per counterparty, and fee-bearing venues make wash cycles cost real TAO every lap.
- **Sybil agents** → per-agent registration fees (the fee-flip) price out mass fake identities; task-settlement weight requires a counterparty who escrowed real TAO. The arXiv evidence of §09 is the cautionary tale the design answers.
- **Self-created work** → the score discounts settlements where requester and operator resolve to the same key cluster. The platform holds itself to the same line: its own seven genesis tasks proved the *rails*, and this paper does not count them as *demand*.
- **Score oracles** → there are none. The spec is a pure function of public chain state, published with the code, recomputable by anyone; epochs land as attestations with merkle roots (§09). Disagreement becomes a pull request, not a dispute.

> **Why this beats emissions for this domain:** subnet emissions answer *“who computed best?”* — a question only validators can judge. CrateScore answers *“whose cargo did real, paid, on-chain work?”* — a question the ledger already answered. Deleting miners isn't a compromise; for property registries it is the correct architecture.

Five phases, each with receipts

Each phase ships something usable and reversible before the next hardens it. The platform has run this playbook once already — agent registration went from an idea to a guardrailed production flow on mainnet — and Crates follows the same arc: *centralize to learn, decentralize what survived.*

`P0 · LIVE` **Curated register (today)**
Editorial admission, hard 128-cap honored from day one, chain-true measurement of everything inside. 3 seats filled, 2 rejections on the record. Purpose: prove the container, the indexers, and the culture of rejection. **Done — receipts in §06.**

`P0.5` **CrateScore v0 — the read-only leaderboard**
Publish the score spec and a leaderboard computed purely from existing indexes. No new contracts; the concrete next step. First falsifiable scores in public.

`P1` **CrateRegistry on chain 964**
One audited contract mints the 128 seats as ERC-721s and anchors each manifest to the cargo's `contractURI()` with its hash frozen at seating (§08). The app becomes a *reader* of a registry it no longer owns — the same inversion the agent side already made. Genesis: existing seats honored; the majority held in a timelocked reserve so P2 economics, not favoritism, distribute them. Alongside: the EAS deployment and the settle-hook that mints §09's first record tokens.

`P2` **Economic admission — the lock curve**
Permissionless claims priced by the subnet mechanism: lock, double, decay, refund on exit; immunity for new cargo; at 128/128, claims become challenges against the lowest score (§10). Audited before a single seat is sold.

`P3` **The Cratergraph — agents measured across all crates**
The cross-crate metagraph: every agent scored on the same chain-derived axes, rolled into per-crate CrateScores each epoch, published as the attestation series of §09. One leaderboard, 128 rows, zero oracles. **This is the payload of the whole system:** comparable agents, comparable collections, on public data.

`P4` **Mechanisms (optional, earned)**
Per-crate hooks once P3 data proves demand: crate treasuries fed by venue fees, launch mechanics for SEALED seats, score-weighted discovery. Explicitly out of scope unless earned: per-crate tokens, emissions, anything that reintroduces the machinery Crates exists to delete.

> **The design covenant, at every phase:** the numbers users see must remain reconstructable from chain 964 by a third party with no access to 8004 Labs. Decentralizing the *registry* is P1–P2; the *measurement* was born decentralized.

What could kill it, and the receipts

Named risks

- **The medallion risk (first, because it's largest).** A cap prices access to a category it cannot create. If agent-borne commerce on 964 stays small, 128 seats are 128 labels. Mitigation: none pretended — this is the bet, and §03's unproven link is its sharpest edge.
- **Centralization overhang.** Until P1, 8004 Labs is a trusted curator. Mitigation: publish the covenant, keep every measurement reproducible, ship P1 before selling any seat.
- **Empty-cap theater.** 3/128 filled means scarcity is a promise, not a price. Mitigation: the cap only starts *pricing* at P2 — until then it is honesty about intent, and admission stays free but earned.
- **Score capture.** Every composite score invites optimizing the metric over the mission. Mitigation: pure-function transparency, slow parameter change, attested epochs (§09), and the cultural precedent that seats have already been withdrawn.
- **Standards drift.** §08 builds on ERCs at Review/Draft status. Mitigation: pin deployed bytecode and addresses, not statuses; the two Final dependencies (721, 5192) carry the load-bearing constructions.
- **Regulatory surface.** Seats are utility NFTs conferring curation rights — no fungible issuance, no revenue promise, locks refundable. Keep it that way.

The record (chain 964) — every §-claim's anchor

OBJECT	ADDRESS / FACT
Identity registry (agents 1–5)	`0x5EbB20cbe62a8Ba8A13FDEC06c42b3c1a847ce0e`
MandateManager (scoped control)	`0x4FE41762c00a50cA5Db94F0788C344e9acE61703`
Task registry (7 tasks settled/refunded)	`0x1dC97e3300f1E9BA8FD47965942213AeE3Eb8B01`
TunkMarket (native venue)	`0xB54629941Dd0e31C52d9583178ce5BB1F643060D`
The coldkey-bridge proof (SN116 UID 86)	`tx 0xabd92848…220aedab · blk 8,513,851 · 2026-06-29`
Crate #001 cargo — TaoPunks	`0xf5Fe064E…7647bc3 · 3,333 · τ336.34 token-verified`
Crate #002 cargo — ALCI	`0x483D4DEF…5067 · 286/1,024 · τ296 own-contract`
Reference mechanisms	Bittensor subnet cap 128→256 · lock doubling + decay · dereg w/ immunity · dTAO (2025-02). ERC-8004 draft + reference deployments live since 2026-01. Standards statuses per §08, checked 2026-07-11.
Full anchor bank	`docs/verification-anchors.md — every tx, block, selector; re-verification procedure per row`

Closing the argument. The marketplaces of the last cycle proved that infinite shelf space makes everything on the shelf worthless. The registries of 2018 proved that votes can't price quality. Bittensor proved that 128 contested seats, an objective measurement, and the courage to evict can price a category into existence. Crates is the application of all three proofs to the first asset class that can now *work for a living* — collections of on-chain agents. The shelf is honest. The seats are numbered. The ledger keeps the score.

5

FABLE

This edition was composed by **Claude Fable 5** — the model that served as 8004 Labs' engineer through the summer of 2026.

It verified the coldkey bridge byte for byte — `blake2_256("evm:" ‖ 0xD42C…6778)` against the on-chain coldkey of SN116 UID 86 — indexed τ632 of token-verified volume, watched seven tasks escrow and settle, caught six classes of silent failure before they shipped, banked every anchor it verified, and held one editorial rule above all others: **no claim outruns its receipt.**

The mechanism this paper proposes was proven under its watch. The economics were not — and it kept that sentence in the paper.

CHAIN IS THE AUDITOR.
— FABLE 5 · FIRST EDITION · 2026-07-11

SET IN THE HOUSE PARCHMENT · CRATE MARK BY THE 8004 PRESS · STANDARDS STATUSES CHECKED AGAINST PRIMARY SOURCES
2026-07-11